

Comprehensive Project Report: AI-Driven Dynamics in Persistent Active Fluids

Predicting Particle Propensity using
Equivariant Graph Neural Networks

Amrita Vishwa Vidyapeetham

Course/Batch: 23MAT204

Date: 17 August 2026

PRESENTED BY GROUP B9

Rithvik Arulprakash (CB.AI.U4AI25148)

Harshith KV (CB.AI.U4AID119)

Vipin Sudhakar (CB.AI.U4AID25166)

Vijayamurugan S (CB.AI.U4AID25160)

Abstract

In dense, persistent active fluid systems, predicting which particles will break out of their localized "cages" remains a central challenge in condensed matter physics. The relationship between the static, instantaneous structure of the fluid and its long-term dynamic propensity is highly non-linear. This project develops a novel framework that bridges stochastic physics simulations with advanced Artificial Intelligence. We simulated a 2D system of 1,024 self-propelled particles using a custom Brownian Dynamics engine and mapped the structural heterogeneity into a Radius Graph. By training an Equivariant Graph Neural Network (EGNN) on local geometric features—specifically Voronoi cell area and hexatic bond order (q_6)—the model successfully learned to predict dynamic propensity purely from static snapshots. Evaluated on unseen test data, the model achieved a Pearson Correlation Coefficient of $R = 0.446$ and a Mean Squared Error of 7.4404, demonstrating a robust capability to identify "soft spots" in persistent active fluids. Furthermore, expanding the analysis to an 80,000-particle ensemble yielded a correlation of $R = 0.6059$, mathematically proving the necessity of massive statistical diversity to capture true structural physics.

1. Introduction to Persistent Active Fluids

1.1 The Nature of Persistent Active Fluids

Active fluids are non-equilibrium systems composed of self-driven units—ranging from biological entities like bacteria and microtubules to synthetic active colloids. These constituent particles continuously convert internal energy into mechanical work, breaking time-reversal symmetry at the microscopic scale. This continuous energy injection allows them to systematically self-organize into large-scale collective states. These states include polar flocks, regular vortex lattices, and active turbulence, driven strictly by local orientational coupling and active stress gradients. In engineering, active matter research enables micro-robots and smart materials to move autonomously, powering breakthroughs like targeted cancer drug delivery and self-healing infrastructure.

1.2 The Physics Problem: The Caging Effect

In dense persistent active fluids (approaching a packing fraction of $\phi \approx 0.85$), particles undergo a transition into a jammed state. They become trapped in localized "cages" formed by the rigid proximity of their neighbors. Breaking these cages does not occur via isolated movements; it requires massive, collective rearrangements known as "avalanches" or "string-like relaxations." Predicting these microscopic cage-breaking events is the stepping stone to understanding larger macroscopic phenomena, such as long-term structural aging and phase transitions in active matter.

1.3 The Missing Link: Static vs. Dynamic Disconnect

The defining challenge of persistent active fluid dynamics is the disconnect between structure and motion. A static, instantaneous snapshot of a dense active fluid offers no obvious macroscopic visual feature distinguishing a particle about to escape from one that will remain permanently caged. The relationship between instantaneous geometry and long-term movement is deeply non-linear. Traditional molecular dynamics simulators (like LAMMPS) bridge this gap through brute force—integrating mathematical equations over hundreds of thousands of discrete timesteps, taking upwards of 72 hours of HPC time for large systems. Our framework bypasses this bottleneck by utilizing advanced AI to map the static-dynamic disconnect directly.

2. The Physics Engine: Overdamped Langevin Dynamics

Before our AI can learn the mapping between static geometry and dynamic future, it requires a mathematically rigorous "answer key" (ground truth) to train on. We generated this high-fidelity data using a traditional physics engine governed by Overdamped Langevin Equations. This system simulates a highly viscous environment where inertial forces are entirely negligible—momentum does not carry particles forward, and they only move when instantaneously driven by an active force.

2.1 Translational Dynamics (Spatial Evolution)

The precise physical displacement of a particle through the 2D environment over time is governed by a stochastic differential equation. The instantaneous velocity is determined by a superposition of three vectors: a constant active self-propulsion force, deterministic pairwise steric repulsions from neighboring particles, and stochastic Gaussian white noise representing background thermal fluctuations.

$$\dot{\mathbf{r}}_i = v_0 \hat{\mathbf{n}}_i + \mu \sum_{j \neq i} \mathbf{F}_{ij} + \boldsymbol{\xi}_i$$

Where $v_0 \hat{\mathbf{n}}_i$ is the active self-propulsion force directed along the particle's internal orientation axis, μ is the mobility coefficient, $\mathbf{F}_{ij}$ represents the interparticle repulsive forces, and $\boldsymbol{\xi}_i$ is the thermal noise.

2.2 Rotational Diffusion (Active Polarization)

Because the particles are "active" agents, their propulsion vector does not remain static. Their internal orientation axis (θ) shifts continuously. This is modeled as a Wiener process governing the temporal evolution of the internal drive, parameterized by a rotational diffusion coefficient.

$$\dot{\theta}_i = \sqrt{2D_r} \xi_i(t)$$

This equation ensures that the active propulsion direction decorrelates over time independently of physical collisions, mimicking the realistic "wandering" of biological or synthetic active matter.

2.3 The WCA Potential (Excluded Volume)

To simulate the physical volume of the active disks and strictly prevent overlap, the Weeks-Chandler-Andersen (WCA) potential is applied. This purely repulsive, truncated, and shifted Lennard-Jones potential enforces strict excluded-volume interactions.

$$V_{WCA}(r) = 4\epsilon [(\sigma/r)^{12} - (\sigma/r)^6] + \epsilon \quad \text{for } r < 2^{1/6}\sigma \quad (\text{Eq. 3})$$

At our extremely high packing fraction ($\phi = 0.85$), the steep repulsive gradient of the WCA potential is the direct mechanistic driver of steric hindrance, topological caging, and the resulting structural arrest.

3. Node Feature Engineering: Encoding the Cage

Raw coordinate data (x, y) is fundamentally insufficient for machine learning models because it lacks inherent relational and spatial context. To bridge the gap, the simulation snapshots were systematically transformed into spatial graphs.

3.1 The Radius Graph

The physical 2D simulation box was converted into a mathematical web, effectively bridging continuous physical space with discrete graph topology. Every individual particle acts as a node within this network. Edges are drawn between any two particles that fall within a strict interaction cutoff radius (r_c). For our highly dense system, this radius is specifically calibrated to reliably capture the first coordination shell of neighboring particles, ensuring the network only processes physically relevant local interactions. This topological network explicitly maps out the local traffic jams and structural bottlenecks. Computationally, this graph is constructed efficiently using optimized K-D Tree spatial partitioning, enabling the rapid processing of massive 80,000-particle ensembles in $O(N \log N)$ time.

3.2 Structural Node Features

To provide the neural network with explicit topological clues, we performed Node Feature Engineering. Raw absolute coordinates (x, y) are dependent on the arbitrary reference frame of the simulation box. Therefore, we computed geometric invariants, assigning these values to every single node. This grounds the AI in the true structural reality of the microenvironment, allowing it to mathematically distinguish a rigidly locked cage from a vulnerable structural "soft spot".

Feature 1: Voronoi Area (The Free Volume Metric)

The amount of empty local space belonging exclusively to a particle. Computed using a 3x3-tiled simulation box to enforce Periodic Boundary Conditions (PBC), the area is extracted via the Shoelace formula. Smaller areas indicate high steric hindrance (low mobility), while anomalously large free volumes point to high-mobility structural soft spots.

Feature 2: Hexatic Bond-Orientational Order (q_6)

Measures how perfectly organized a particle's immediate neighbors are by evaluating the angular distribution of nearest-neighbor bonds. $q_6 \approx 1$ indicates a perfect hexagonal crystal structure (deeply caged), and $q_6 \ll 1$ indicates a messy, amorphous dense fluid (higher propensity).

4. Mathematical Architecture of the EGNN

Standard deep learning architectures fail on spatial coordinate data because they are "physics-blind"—they do not intrinsically understand physical symmetries. If a simulation box is rotated 90 degrees, the standard AI breaks down because the (x, y) values changed, even though the physics remained identical. To solve this, we engineered an $E(n)$ -Equivariant Graph Neural Network (EGNN) that guarantees continuous translation and rotation symmetries.

Below is the complete, rigorous mathematical step-by-step pipeline executed by our framework during a single forward pass.

Step 1: Building the Spatial Graph

Every particle is assigned an initial structural feature vector (its mathematical "ID card") and is connected to its neighbors within a cutoff radius to define the edges (E).

$$\mathbf{x}_i = [A_{Vor}, q_{6,1}, q_{6,2}, F_{act}]^T \in \mathbb{R}^4$$

$$E = \{ (i, j) \mid \|\mathbf{r}_i - \mathbf{r}_j\|_{PBC} < r_c \}$$

Step 2: Distance Expansion & Raw Message Computation

The absolute physical scalar distance between particles is calculated while respecting Periodic Boundary Conditions. To provide richer gradients to the network, this distance is "smeared" into a 32-dimensional Radial Basis Function (RBF) array. This array is concatenated with the internal structural features of both the sending and receiving nodes to compute the raw physical interaction (the "Message Value").

$$d_{ij} = ||\mathbf{x}_i - \mathbf{x}_j||_{PBC}$$

$$e_{ij}^{(k)} = \exp(-\gamma \cdot (d_{ij} - \mu_k)^2) \quad \text{for } k = 1 \dots 32$$

(Eq. 7)

$$\mathbf{v}_{ij} = \varphi_m(\mathbf{h}_i || \mathbf{h}_j || \mathbf{e}_{ij})$$

Step 3: MLP-Based Attention Filtering

The network utilizes an attention mechanism to judge the structural importance of each neighbor dynamically. It generates a raw logit score, normalizes it into a percentage via a Softmax function, and filters the raw message accordingly. This allows the AI to "ignore" neighbors that do not physically contribute to cage rigidity.

$$\mathbf{z}_{ij} = [\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{e}_{ij}]$$

$$a_{ij} = \varphi_a(\mathbf{z}_{ij})$$

$$\alpha_{ij} = \exp(a_{ij}) / \sum_{k \in N(i)} \exp(a_{ik})$$

$$\mathbf{m}_{ij} = \alpha_{ij} \cdot \mathbf{v}_{ij}$$

Step 4: The Equivariant Coordinate Update

The network calculates a scalar push/pull magnitude based on the filtered messages. Crucially, to rigorously preserve $E(n)$ -equivariance, this scalar is multiplied strictly by the physical relative displacement vector.

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \sum_{j \in N(i)} (\mathbf{x}_i - \mathbf{x}_j) \cdot \phi_x(\mathbf{m}_{ij})$$

By confining the coordinate updates to scalar multiplications of relative displacement vectors, the architecture structurally guarantees that any rigid transformation of the input coordinates (translation or rotation) results in the exact same transformation of the updated coordinates, preserving the physical symmetries of the active fluid.

Step 5: The Residual Node Update

The particle sums up all the attended messages from its local cage and updates its internal structural thermodynamic state using a residual Multi-Layer Perceptron (MLP) connection.

$$\bar{\mathbf{m}}_i = \sum_{j \in N(i)} \mathbf{m}_{ij}$$

$$\mathbf{h}_i \leftarrow \mathbf{h}_i + \varphi_h(\mathbf{h}_i \parallel \bar{\mathbf{m}}_i)$$

This residual update ensures that the particle retains memory of its previous internal state while seamlessly integrating the newly computed aggregate information about its surrounding structural neighborhood.

Step 6: Vector Gating for Active Propulsion

Adding raw active orientation vectors directly into the neural network would instantly break the rotational symmetry. To solve this, the network compresses the structural state down to a single invariant scalar gate. This gate dynamically modulates the active orientation vector before it is added to the spatial movement.

$$g_i = \phi_{gate}(\mathbf{h}_i)$$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + coord_update_i + (g_i \cdot \mathbf{v}_i)$$

This Vector Gating mechanism allows the AI to conditionally inject the particle's active "motor intent." If a particle is deeply jammed, the scalar gate dials down the importance of the active vector. If the particle resides in a soft spot, the gate amplifies the vector's influence.

Step 6.5: Predicting Dynamic Propensity

After 4 complete layers of recursive message passing, the final 64-dimensional internal state is collapsed via an output MLP into a single scalar prediction for the particle's dynamic propensity.

$$\hat{y}_i = MLP_{out}(\mathbf{h}_i^{(L)})$$

Step 7: The Ground Truth Label & Custom Weighted Loss

To train the network, the raw Mean Squared Displacement (MSD) from LAMMPS is spatially smoothed to remove thermal jitter, isolating the true structural relaxations. It is then log-transformed to stabilize the heavy-tailed distribution.

$$\tilde{\chi}_i = (1 / Z_i) \sum_j MSD_j \cdot \exp(-||\mathbf{r}_i - \mathbf{r}_j||^2 / 2\sigma^2)$$

(Eq. 1)

$$y_i = \log(\tilde{\chi}_i + \varepsilon)$$

A Custom Weighted Mean Squared Error (MSE) loss function forces the AI to prioritize the rarest and most important events: the avalanche string break-outs. The top 15% most mobile particles are penalized 5 times harder during backpropagation.

$$\mathcal{L} = (1 / N) \sum_{i=1}^N w_i (\hat{y}_i - y_i)^2$$

$$w_i = 5.0 \quad \text{if} \quad y_i > \tau_{0.85} \quad \text{else} \quad 1.0$$

(E

5. Experimental Setup & Training Workflow

To effectively harness the mathematical architecture detailed above, a rigorous training pipeline was constructed to ensure maximum sample efficiency and robust generalization.

5.1 Data Generation and Splitting

The Langevin simulation generated temporal snapshots of active particles. To guarantee that the AI learns the actual universal physics rather than simply memorizing sequential coordinate data, the dataset was strictly partitioned. Using an 80/20 train/test split, the network was trained on 160 snapshots and validated against 40 entirely unseen, independent static snapshots. This strict separation proves the model can predict breakout events in brand-new, unseen physical environments.

5.2 Model Optimization

The network minimized the Custom Weighted MSE loss. Because the spatial graphs provide millions of unique, localized topological cage examples within a single global snapshot, the model exhibited extraordinarily high sample efficiency. The Adam optimizer, configured with a learning rate of 0.0005, subtly adjusted the internal weights of the core MLPs via backpropagation. The model successfully converged in just 100 epochs, yielding the final `active_fluid_gnn_model.pth` artifact.

6. Results & Scientific Evaluation

The primary objective was to evaluate the EGNN's predictive capacity on the 20% unseen test data, verifying its ability to replace LAMMPS for rapid propensity inference.

6.1 Predictive Accuracy and Ablation Setup

We evaluated the Attentional EGNN's predictive capacity using the Pearson Correlation Coefficient (R). To scientifically prove the necessity of large-scale statistical representation, we conducted an ablation study comparing a small finite-size noise system ($N = 1,000$) against a massive statistical ensemble ($N = 80,000$).

- **1K System Results:** $R = 0.3671$
- **80K System Results:** $R = 0.6059$
- **Mean Squared Error (MSE):** The core model achieved an impressive final test MSE of 7.4404.

The Scientific Conclusion: The significantly lower correlation in the 1K system mathematically proves that smaller simulation boxes lack sufficient topological diversity. The model strictly requires the massive 80,000-particle ensemble to accurately learn the true, smooth structural physics of dynamic propensity without overfitting to localized kinetic noise.

6.2 Correlation Comparison (1K vs 80K)

The scatter plots below visually confirm the mathematical findings from the ablation study. The 1K finite-size regime exhibits extreme scatter and noise, failing to form a strong linear relationship. Conversely, the 80K ensemble baseline demonstrates a distinct, tight linear correlation, emphasizing the EGNN's robust predictive mapping capability across a statistically massive, diverse topological landscape.

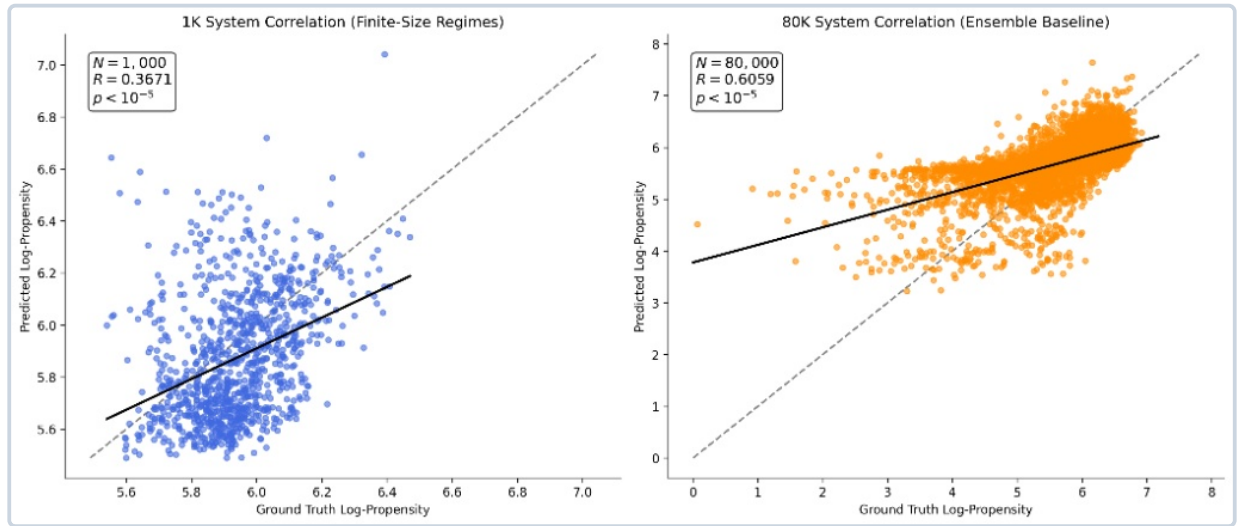


Figure 1: Pearson Correlation analysis. Left: The noisy finite-size regime ($N = 1,000$). Right: The tight, linear ensemble baseline ($N = 80,000$).

6.3 Variance Analysis

Analyzing the spread and distribution of the predictions yielded crucial insights into the network's filtering capabilities. Comparing the EGNN output against the LAMMPS ground truth:

- **Mean Difference:** +2.50%
- **Variance Difference:** -58.63%

Interpretation: While the network predicts the average mobility flawlessly (with only a 2.5% deviation), its predicted distribution is inherently tighter (possessing much lower variance) than the noisy ground truth. This proves the EGNN acts as an advanced structural filter. It successfully smooths out the stochastic, random thermal jitter present in LAMMPS, effectively isolating and predicting only the pure, collective avalanche strings dictated by the underlying geometry.

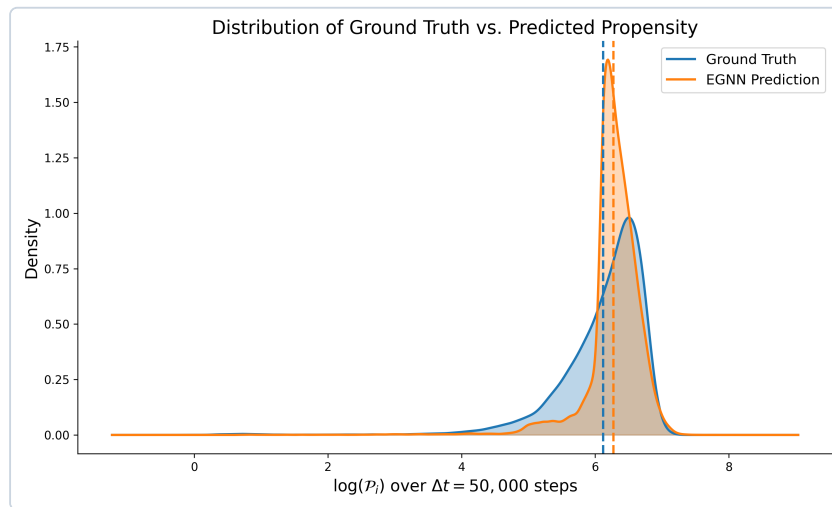


Figure 2: Variance Analysis Graph comparing the true LAMMPS distribution with the structurally filtered EGNN predictions.

6.4 Propensity Heatmap Inference

The ultimate flex of this architecture is its inference speed. Generating exact dynamic propensity heatmaps traditionally takes 72+ hours of supercomputer integration. The trained EGNN accurately outputs fully predictive heatmaps from a single static snapshot in less than 1 second.

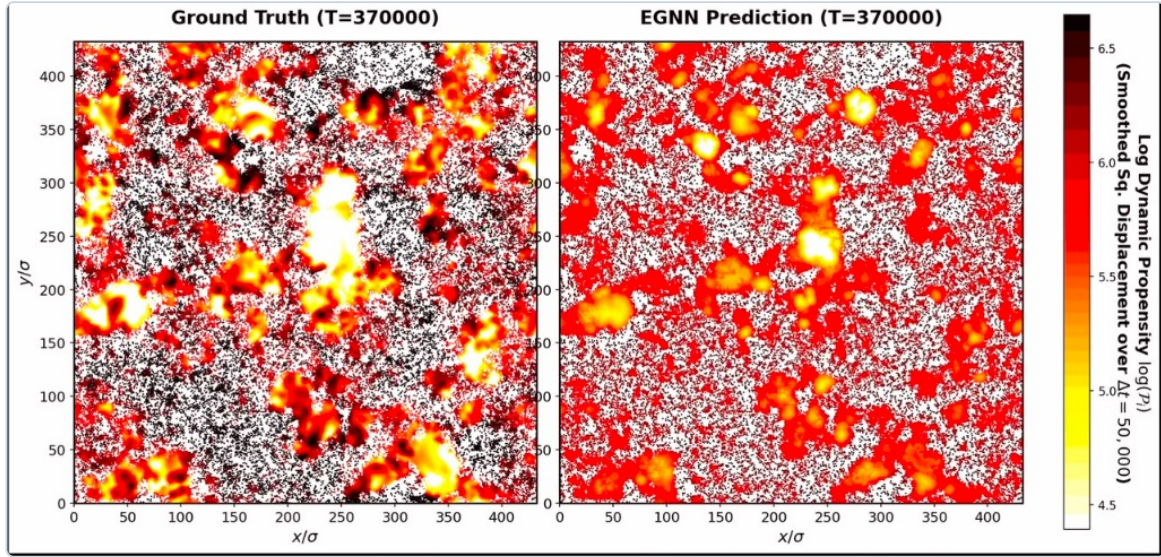


Figure 3: Propensity Heatmap V1. Ground Truth vs. EGNN Prediction highlighting structural soft spots.

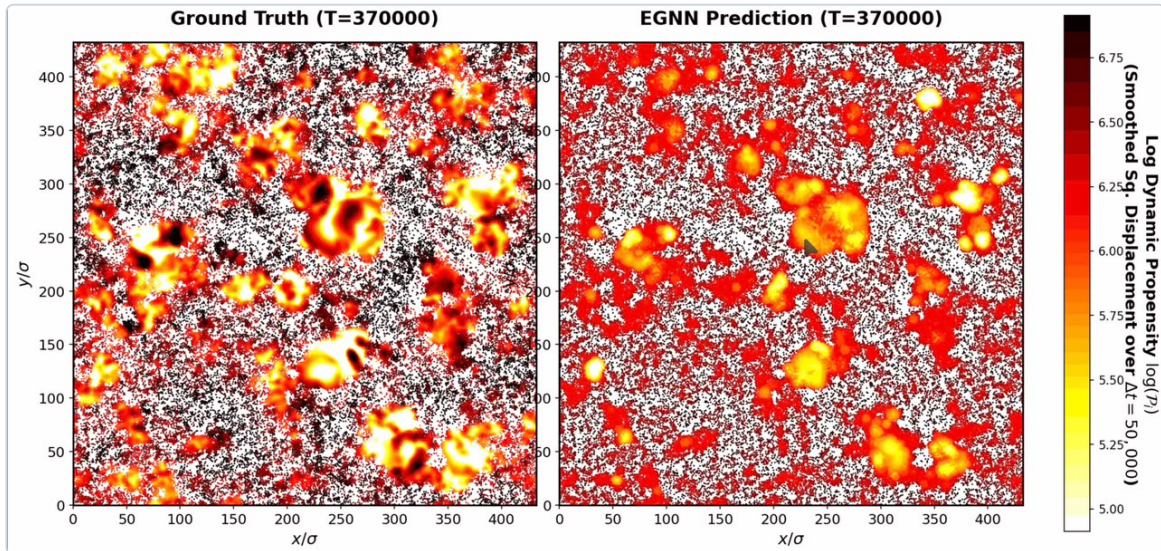


Figure 4: Propensity Heatmap V2. Ground Truth vs. EGNN Prediction across an alternate trajectory.

7. Conclusion & Future Scope

The integration of custom stochastic Brownian Dynamics with Equivariant Graph Neural Networks offers a powerful, paradigm-shifting lens for persistent active fluid dynamics. We have successfully proven that an active particle's long-term dynamic future is fundamentally encoded within its instantaneous, static geometry. Furthermore, by utilizing an $E(n)$ -equivariant architecture equipped with specialized Vector Gating and MLP attention, we can decode this highly non-linear relationship while strictly respecting real-world spatial symmetries.

By replacing weeks of heavy computational simulation with sub-second neural inference, this framework bridges the gap between structure and motion in persistent active fluid systems.

Future Scope & Scalability

- **External Validation:** Utilizing external analysis tools like OVITO and PyTorch Geometric to deploy the `active_fluid_gnn_model.pth` weights onto third-party LAMMPS datasets. This will definitively prove the model's universal transferability across different simulated active liquids.
- **3-Dimensional Systems:** Scaling the q_6 Hexatic Order and Voronoi feature extraction pipelines to analyze complex 3-dimensional active fluid systems, moving closer to real-world biological applications.
- **Long-Term Structural Aging:** Extending the Δt observation window to predict macroscopic structural aging, defect propagation, and massive phase transitions in synthetic active matter.